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Build an iCE40 PWM Demonstration

RTL, mapped simulation, routed timing and bitstream checks; physical deployment remains open

A candidate iCEstick design cycles through five exact PWM duty levels. Independent RTL and mapped-netlist simulations check a full production sequence. Managed synthesis and routing produce an actual packed configuration, which is decoded and checked. Behavioral, timing and configuration mutations test the guards. No board or programmer was available: programming and physical observations remain required before this demonstration is complete.

SECTION 01

Software evidence and the open hardware gate

The **software build is verified** for a candidate iCEstick HX1K target. The original physical-deployment requirement remains incomplete because no board or programmer is available.

CHECK	OBSERVED RESULT	STATE
RTL and mapped top	5,243,400 cycles each; all five duty stages	PASS
Independent comparisons	5,263,881 per production top run	PASS
Core boundary/reset tests	Two parameter configurations; 10,155 checks each	PASS
Final route timing	191.35 MHz estimate; 12 MHz target	PASS
Packed configuration	32,220 bytes; decoded configuration matches	PASS
Negative controls	Duty, impossible timing and changed bit rejected	PASS
Physical deployment	No programming or measured board behavior	OPEN

Crucible is beta. It supplies the managed Yosys, Icarus Verilog, nextpnr and IceStorm environment. These tools retain their authors' credit and licences. No vendor or tool-project endorsement is implied.

What the route result means

The reported frequency is a **nextpnr device-model estimate for the final routed design**. It is not an observed oscillator frequency, a board measurement or vendor signoff.

SECTION 02

Freeze the target and the PWM contract

The candidate target is **Lattice iCEstick, iCE40HX1K in TQ144**. The manufacturer guide and nextpnr's example agree on the clock and red LED pin mapping.

ITEM	SPECIFIED VALUE
Clock input	12 MHz nominal oscillator; package pin 21
LED output	D1 red LED; active high; package pin 99
PWM period	256 clock samples
High samples per period	0, 64, 128, 192 and 256, then repeat
Hold time per stage	4,096 complete PWM periods
Startup reset	Seven synchronous clock edges after configuration

An eight-bit phase counter advances each clock. A period counter advances when phase wraps; the duty stage advances after 4,096 periods. The output compares the phase with the nine-bit duty value, allowing both exact 0% and exact 100% endpoints. Stage updates occur on a period boundary.

The top initializes a three-bit startup register to zero using the iCE40 configuration model. The engine state resets synchronously while startup is below seven; its LED output is forced low while reset is high. There is no unassigned external reset input.

At the nominal clock, the derived PWM carrier is **46.875 kHz**, each duty stage lasts **87.381 ms**, and the five-stage sequence lasts **436.907 ms**. These are arithmetic predictions, not measurements from a physical board.

> **Pin-direction constraint**
> Package pin 21 remains an **input**. The manufacturer warns against assigning an output to the oscillator connection. The supplied PCF constrains both top-level ports.

SECTION 03

Compare against an independent cycle model

The observer computes expected behavior from **elapsed clock counts**, without reading DUT phase, stage, duty or period-counter state.

TEST	MEASURED SCOPE
Core, one period per stage	10,120 cycles; 10,155 comparisons
Core, two periods per stage	10,120 cycles; 10,155 comparisons
Production RTL top	5,243,400 cycles; 5,263,881 comparisons
Production mapped top	Same complete testbench and default configuration
Complete periods per top run	20,481 checked; all five stages and wrap
Per-stage complete periods	4,097 at zero; 4,096 at each other duty

Core tests exercise synchronous reset during activity and one edge before a period boundary, exact endpoints, stage transitions and sequence wrap. The production test keeps **4,096 periods per stage** and traverses the entire five-stage sequence; the mapped test does not use a faster replacement counter.

Every post-edge output is checked with four-state comparison. Whole-period high samples must equal the expected duty. The zero-duty period count includes the additional checked period after the sequence wraps.

Mapped simulation uses **unchanged official iCE40 cell models from the pinned Yosys artifact**, copied locally and hashed. The final observer binds each result to current source, testbench, model and stage inputs. These are zero-delay functional tests; they do not measure physical glitches or path delays.

SECTION 04

Synthesize, place and route the actual design

Yosys maps the production top to iCE40 primitives. nextpnr targets **HX1K / TQ144**, fixes the PCF pins, uses seed 1 and constrains the only clock to 12 MHz.

RESOURCE	FINAL ROUTE USE
Logic cells	47 / 1,280
User I/O	2 / 112
Global buffers	2 / 8
Block RAM	0 / 16
PLL	0 / 1

TIMING ITEM	REPORTED VALUE
Clock constraint	12.00 MHz
Final estimated maximum frequency	191.35 MHz
Positive route verdict	PASS; process exits zero
Impossible target	1,000 MHz rejected; process exits one

The JSON synthesis result contains **39 LUT4s, 20 carry cells and 26 flip-flops** before nextpnr packing. A disconnected, zero-port Yosys scope annotation is metadata. Unknown unmapped logic is rejected by the observer.

The final routed frequency is reported separately from the earlier placement estimate. The LED is an asynchronous external observation point with no external receiving-clock requirement; this experiment establishes internal clock closure in the open device model. It is not a complete board timing or electrical qualification.

SECTION 05

Decode the real packed configuration

IceStorm packs the routed ASC into an actual **32,220-byte bitstream**. The check decodes that bitstream and compares configuration records with the route output.

CHECK	OBSERVATION
Logic and I/O configuration	Every configuration record agrees
Extra configuration bits	Retained in the comparison
RAM initialization	Omitted source records expand to zero
Normalized record count	265 records match
Unpack and repack	Packed bytes are identical
Changed configuration control	One logic bit survives packing and is rejected

Comments and symbol labels do not encode FPGA configuration and are excluded. The source ASC omits unused RAM initialization; iceunpack writes sixteen zero-filled RAM records. IceStorm's source confirms the zero-initialization behavior. Normalization inserts those exact zero defaults, retaining all actual RAM bits.

A one-bit mutation changes logic tile (1,1), row 0, column 0 in a scratch ASC. That configuration is packed and decoded through the real tools. The same record comparator rejects it. A nonzero RAM-data mutation also differs from the normalized positive configuration.

> **Integrity boundary**
> Configuration agreement and byte-identical repacking establish **serialization consistency**. They do not prove post-route logical equivalence, physical loading, programming success or correct behavior on silicon.

SECTION 06

Make each negative result discriminating

A negative control passes only when the **intended defect is detected**. Tool setup errors, missing binaries and frontend failures do not satisfy these checks.

CONTROL	REQUIRED DETECTION
Broken duty endpoint	At phase zero, duty zero: observed 1, expected 0
Impossible clock	Final timing reports FAIL at 1,000 MHz
Changed configuration bit	Decoded records differ from the positive route

```
PWM_MISMATCH cycle=256 elapsed=256 phase=0
stage=0 duty=0 got=1 expected=0
ERROR: Max frequency ... FAIL at 1000.00 MHz
PWM_CONFIGURATION_MISMATCH after bitstream decoding
```

The duty fixture uses an inclusive comparison, creating a high sample at the nominal zero endpoint. It fails the same independent core observer that passes the corrected implementation. The actual failing test was observed before introducing the corrected comparator.

The impossible-clock run retains the original design, package, PCF and seed. Only the timing target changes. The configuration mutation is made in a separate scratch output; the positive bitstream is preserved. All three negative processes exit one, and their causal diagnostics are inspected.

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SECTION 07

Reproduce and inspect the software evidence

Use the **FPGA_Demo repository root in Linux/WSL x86-64**, with Crucible already installed. Python observers use the standard library.

```
igny init
igny workspace sync
bash scripts/run.sh identity
bash scripts/run.sh rtl
bash scripts/run.sh synth
bash scripts/run.sh mapped
bash scripts/run.sh route
bash scripts/run.sh pack
bash scripts/run.sh negative
python3 scripts/verify.py
```

ARTIFACT	WHAT TO INSPECT
reports/top_rtl.log / top_mapped.log	Full production-cycle comparisons
reports/route.json	Final timing and device utilization
build/fpga.asc / fpga.bin	Actual route and packed configuration
reports/*_inputs.json	Source/configuration hash bindings
reports/summary.json	Bounded measured result and open hardware state

Raw build outputs, managed payloads and tool logs remain local. The public repository retains source, tests, pinned tool metadata, compact evidence and exact reviewed whitepapers. Its README lists prerequisites, all commands, expected results and manual Git steps.

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SECTION 08

Finish physical deployment

The original demonstration requires **real programming and observation**. This remains an open acceptance gate; no connected board was probed or programmed.

REQUIRED EVIDENCE	ACCEPTANCE RECORD
Confirmed physical target	Board model/revision, HX1K/TQ144 and pin review
Host programming access	Actual USB/driver permissions and programmer identity
Exact loaded artifact	Reviewed bitstream hash and programming log
Clock and output capture	Measured clock, PWM period and five duty levels
Startup/reset behavior	Power-cycle/configuration capture and repeatability
Visual observation	D1 response and sequence consistent with captures

The iCEstick guide describes the onboard FTDI programming interface and SPI flash path. Any later programming must first confirm the actual board revision, source pin mapping and selected image. WSL USB access and physical programmer behavior have not been validated here.

The README includes a clearly conditional manual programming command. The automation has **no programming stage**. Once hardware is available, record actual measurements, review failures, and update the public evidence and this document before marking deployment complete.

> **Current full-project state**
>
> **Incomplete: physical board and programmer unavailable.** The verified software package supports that remaining work without replacing it with a simulation claim.

SECTION 09

Limits and primary source notes

All software results were checked locally on **1 November 2026**. The candidate build is specific to the stated device, package, pin constraints, seed and clock model.

No physical clock, LED duty, startup interval, reset waveform, loaded-bitstream identity or programming operation has been observed. The simulations have no SDF; the configuration comparison is not a logical equivalence proof. Other boards, devices, seeds, tool versions, environmental corners and independent clean-machine reproduction remain unverified.

The design has a combinational PWM output, evaluated after synchronous state updates in functional simulation. Hardware pulse integrity and transient behavior require measurement. nextpnr timing is an open device-model estimate and carries no vendor signoff claim. No ASIC layout, foundry qualification or fabricated-silicon result is asserted.

Lattice · EB82 iCEstick Evaluation Kit User Guide
https://www.latticesemi.com/view_document?document_id=50701

YosysHQ nextpnr · iCEstick pin constraints
https://github.com/YosysHQ/nextpnr/blob/main/ice40/examples/blinky/blinky.pcf

YosysHQ IceStorm · ASC/bitstream and RAM defaults
https://github.com/YosysHQ/icestorm/blob/main/icepack/icepack.cc

YosysHQ Yosys · official iCE40 functional cells
https://github.com/YosysHQ/yosys/blob/main/techlibs/ice40/cells_sim.v